OTC-01 | ROTATING EQUIPMENT

Centrifugal Pumps & Mechanical Seal Reliability

SUGGESTED DURATION: 3 DAYS | IN-COMPANY / LIVE ONLINE

Recurring seal leakage often reflects a wider pump-system problem. This programme connects hydraulic duty, installation, operating practice and seal support conditions so teams investigate the mechanism before replacing parts.

WHO SHOULD ATTEND

Maintenance, operations and reliability engineers; pump technicians; supervisors and plant managers.

WHAT PARTICIPANTS WILL BE ABLE TO DO

- Read pump duty, performance curves and the relationship between head, pressure and density.
- Recognise cavitation, low-flow operation, misalignment and seal-support problems.
- Develop an evidence-based action plan to improve pump and seal reliability.

SUGGESTED PROGRAMME OUTLINE

Day 1 | Hydraulic duty and operating behaviour

Head, flow, system resistance and duty-point interpretation; BEP and manufacturer operating limits; NPSH available, NPSH required and margin; suction piping, minimum-flow protection and field measurements.

Day 2 | Mechanical seals and supporting systems

Seal components and leakage mechanisms; materials and service compatibility; single and dual seal concepts; selected API 682 support-plan principles; flush, quench, barrier and buffer arrangements; start-up and shutdown precautions.

Day 3 | Diagnose recurring failures

Alignment, soft foot, runout, bearings and lubrication; interpret vibration, temperature and leakage evidence; distinguish symptom from mechanism; build a pump/seal failure investigation and reliability action plan.

OTC-01 | PRACTICAL APPLICATION & DELIVERY

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PRACTICAL WORKSHOPS

Caustic transfer pump seal failure

Use an illustrative operating history, seal photographs and flush readings to test possible causes; propose safe checks and rank corrective actions.

Pump duty review

Locate the operating point on supplied curves and identify the data needed before recommending throttling, speed change or impeller modification.

PARTICIPANT TOOLS AND TAKEAWAYS

Pump inspection checklist; seal investigation worksheet; operating-data log; prioritised reliability action-plan template.

HOW LEARNING IS DELIVERED AND ASSESSED

Facilitated explanations, worked examples, case discussions and small-group exercises. A short baseline check establishes starting knowledge; a closing knowledge check and practical review assess learning. Assessment can be adapted for operators, technicians or experienced engineers.

CUSTOMISATION AND CLIENT INPUTS

Fluid properties, pump curve, seal arrangement and support schematic, operating history and available failure photographs. Equipment disassembly is optional and requires client facilities and supervision.

The agenda is indicative. Final content, duration, batch size, language and delivery arrangements are agreed after a training-needs discussion. Online delivery uses document reviews, calculations, role-play and case exercises; physical demonstrations require separate site arrangements.

FACILITATOR

Prem Prakash Mishra | Mechanical engineer, advisor and corporate trainer. Brings 39 years of career experience across Kuwait Oil Company, IPCL and NTPC, spanning projects, commissioning, maintenance, reliability, asset integrity and industrial audits. Specialist inputs, where needed, are agreed in the proposal.

COMPLETION AND FOLLOW-UP

An Omtech Corporation participation or completion certificate may be provided under the agreed assessment arrangements. This is company training, not an accredited professional qualification or a guarantee of certification. Post-training support is available with scope and duration agreed in the proposal.

REQUEST A CUSTOMISED TRAINING PROPOSAL

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